

FROM PAST SALES TO FUTURE PURCHASES

Customer Profiles and Promotion Insights

Dunnhumby Complete Journey | Portfolio Project

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PostgreSQL · Python · BG/NBD

Summary:

This project combines historical sales with expected future purchases to create customer profiles, then compares their historical discount and coupon patterns. The analysis shows that past sales are a strong signal of future activity, while forward-looking purchase predictions add useful differentiation for some customers. Promotion patterns provide additional context, but they do not support a simple “more promotion = better future activity” conclusion. The profiles are therefore best used to identify where different promotion, and engagement approaches may be worth exploring further.

Github repository: [lillelilla/Customer-profiles-promotion-insights](https://github.com/lillelilla/Customer-profiles-promotion-insights)

Including the project documentation, SQL workflow, Python notebook, and presentation.

Understanding Customer Profiles Through Past Sales, Future Activity, and Promotions

Dataset: dunnhumby Complete Journey | anonymized grocery retailer | 2,500 source households | 2,498 analyzed households



Project context

The **dunnhumby Complete Journey** dataset contains about two years of grocery purchases for **2,500 frequent-shopper households** at an anonymized retailer. It includes baskets, sales, loyalty-card discounts, manufacturer coupons, and retailer coupon matches.

The retailer operates in a **non-contractual environment**: customers do not formally subscribe or cancel, so a household that stops shopping may be temporarily inactive or may have effectively left.



Business problem

Historical sales show how much a household contributed in the past, but households with similar past sales can still have very different future purchasing activity.

This motivates combining historical sales with expected future basket activity, then examining whether the resulting profiles also differ in historical discount and coupon behavior.



Project objective

Create customer profiles by combining **past sales contribution** with **expected future purchase activity**, then compare how these groups historically received discounts and redeemed coupons. The goal is to identify **meaningful customer differences and loyalty approaches worth testing**.



Core project questions

Can past sales contribution and expected future activity be combined to support customer prioritization, and

How do discount and coupon patterns differ across the resulting customer profiles?



Data & analytical approach

The analysis turns product-level transactions into customer-level measures, uses the BG/NBD model to estimate each household's expected future shopping activity from its historical purchase pattern, and tests those predictions on a later 167-day holdout period before using them to build customer profiles.

1. Data and customer unit

- The [dunnhumby Complete Journey dataset](#) contains transaction data for **2,500 households**.
- Only the transaction data is used in this project. Other available files, such as demographic or campaign-related data, are outside the scope of the analysis.
- The raw transaction data is recorded at the product level, so individual product records are first combined into shopping baskets, with each basket representing one shopping trip. Each household is then treated as one customer.
- The final analysis uses **2,498 households with purchase history before the prediction point**. The remaining 2 households first appeared during the holdout period, so they had no historical purchase data available for modelling and were excluded.

2. What is measured from the historical period?

For each household, the historical period captures three aspects of past behavior:

- **Historical sales:** Total retailer sales generated during the period.
- **Purchase activity:** How often and how recently the household shopped.

Promotion receive/usage: How often baskets included loyalty-card discounts, manufacturer coupons, or retailer coupon matches, plus selected promotion-value measures.

3. How is expected future activity estimated?

A repeat-purchase model called **BG/NBD** (Fader, Hardie and Lee, 2005a, 2005b) estimates how many shopping baskets each household is expected to make over the next **167 days**.

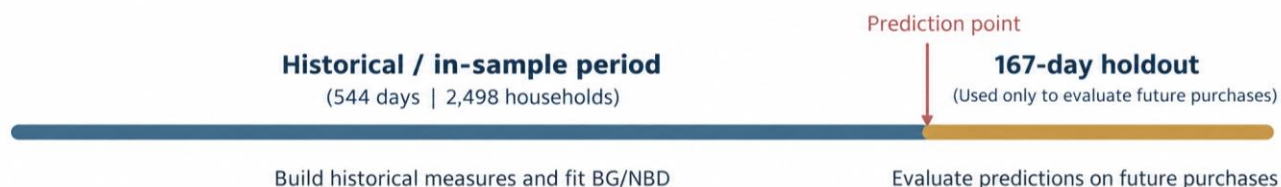
It uses three parts of the household's purchase history:

- **Purchase frequency:** How many repeat shopping trips the household made.
- **Purchase recency:** How much time passed between its first and most recent observed purchase.
- **Observation period :** How long the household had been observed since its first purchase.

The model produces an **expected future basket count** for each household.

To check whether the model adds value, its predictions are compared with a simple benchmark based on the household's **historical purchase rate**.

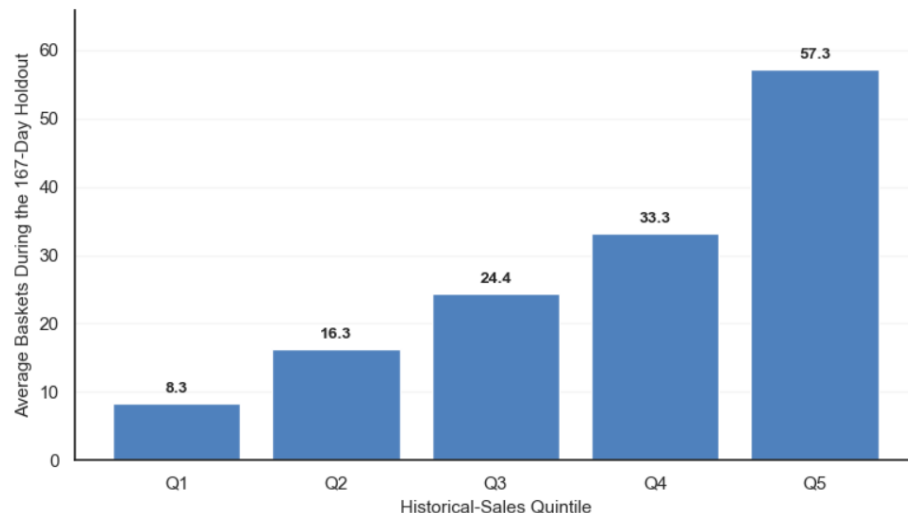
4. Holdout evaluation



Past Sales Provide a Strong Signal; Future-Purchase Predictions Add Further Differentiation

Customers with higher historical sales generally made more purchases later. The next step is to test whether a future-purchase model (BG/NBD) can further help identify, within each historical-sales group, which households are expected to make more or fewer future purchases.

Higher Historical Sales Are Associated with More Future Purchases

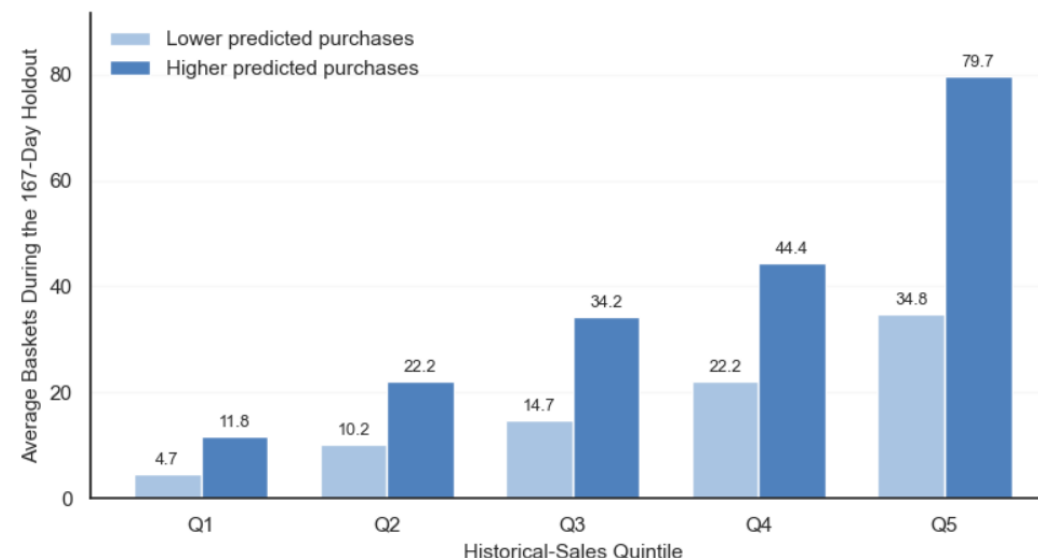


Note: Q1–Q5 divide households into five equally sized groups based on historical sales, from Q1 = lowest sales to Q5 = highest sales.

1. Past sales already provide a strong indication of future purchase activity in this case

Households with higher historical sales generally made more purchases during the 167-day holdout. Average future baskets increased from **8.3 in Q1** to **57.3 in Q5**.

Within Similar Historical-Sales Groups, Households Predicted to Make More Future Purchases Also Made More Actual Purchases



2. BG/NBD additionally shows differences that past sales alone can miss

BG/NBD uses *how often and how recently households shopped* to estimate future purchases. For example, within the same **Q3 past-sales group**:

- Lower predicted purchases → 14.7 actual future baskets
- Higher predicted purchases → 34.2 actual future baskets

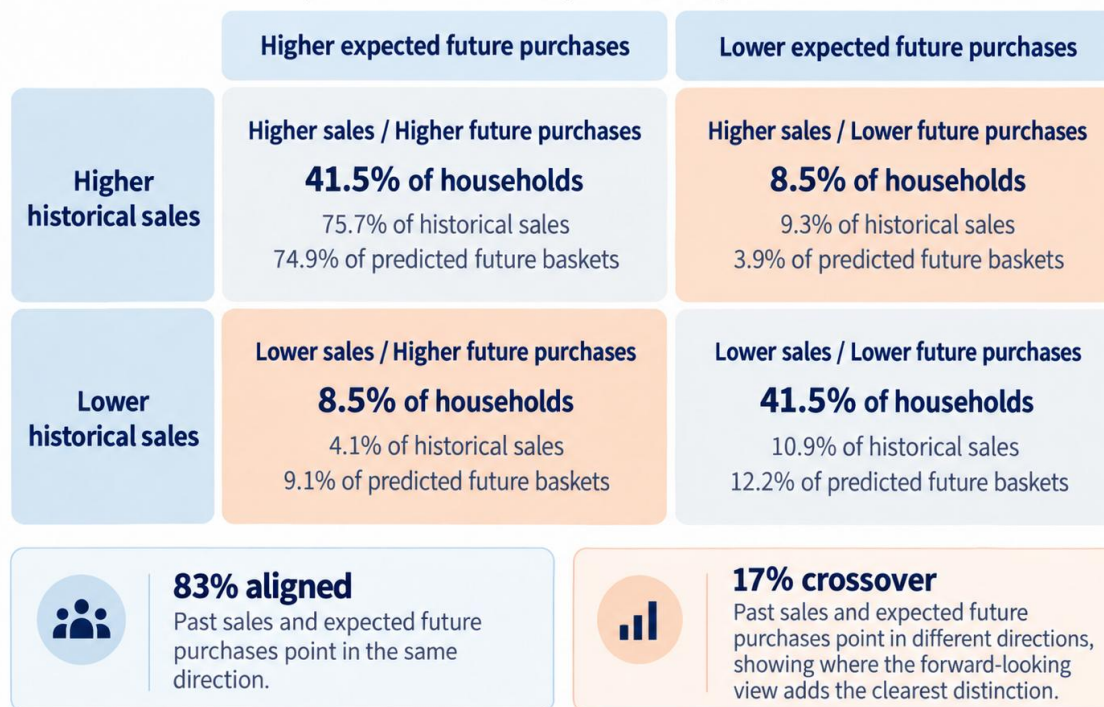
KEY TAKEAWAY: Even customers with similar broad past sales can have very different future purchase activity. Thus, BG/NBD helps distinguish which customers within similar past-sales groups are likely to be more or less active in the future.

Customer Profiles Show Where Past Sales and Future Activity Differ

Customers are grouped using historical sales and expected future purchases. Most customers show a similar pattern on both, while a smaller group has a different outlook than their past sales suggest.

Customer profiles: Historical sales × expected future purchases

The 17% crossover groups (8.5% lower sales / higher future purchases + 8.5% higher sales / lower future purchases) show what a past-sales-only view can miss.



Note: “Higher” and “Lower” are defined relative to customer-base medians: historical sales ≈ **\$1,572** and expected future activity ≈ **18.8 baskets over 167 days**. Because both dimensions are split at their medians, each side contains about 50% of households, which creates the symmetric **41.5% / 8.5% / 8.5% / 41.5%** profile sizes. The important result is therefore the **83% aligned vs. 17% crossover** split, not the symmetry itself.

What the two-dimensional profile framework tells us that historical sales alone would not:

Higher sales / Lower future purchases

- These customers have contributed meaningful sales in the past but may become less active going forward.
- Implication: Priority group for re-engagement testing, because

Lower sales / Higher future purchases

- Lower past contribution, but these customers are expected to shop more often and may have room to increase basket value or category of spending.
- Implication: Opportunity to develop future sales value

Higher sales / Higher future purchases

- This group represents a large share of both historical sales and expected future purchases.
- Implication: Protect and retain

Lower sales / Lower future purchases

- Large customer group, but relatively small contribution on both dimensions.
- Implication: Use scalable, lower-cost engagement and test where extra investment is justified, because broad high-cost incentives may not be justified unless testing shows a clear response.

Promotion context:

Retail Loyalty-Card Discounts Are Widespread; Coupons Are Much Less Common

We first establish how common each promotion type is across the customer base. This gives context for interpreting whether differences in discount and coupon use across the customer profiles are meaningful or simply reflect patterns that are common across the whole customer base.



Note: The chart shows the average household's promotion rate. Each household is weighted equally, regardless of how many baskets it made, because the question is: *For a typical household, what share of its baskets contained this promotion?* Example: if Household A received a retailer discount on 90% of its baskets and Household B on 70%, the average is 80%, even if Household A made more baskets.

Retail discounts are part of normal shopping for most households. The average household received one on 84.2% of baskets.

Manufacturer coupons are less common. The average household redeemed one on 5.7% of baskets, and 31% never redeemed one.

Retail coupon matches are rare. The average household received one on 2.7% of baskets; the median household received none.

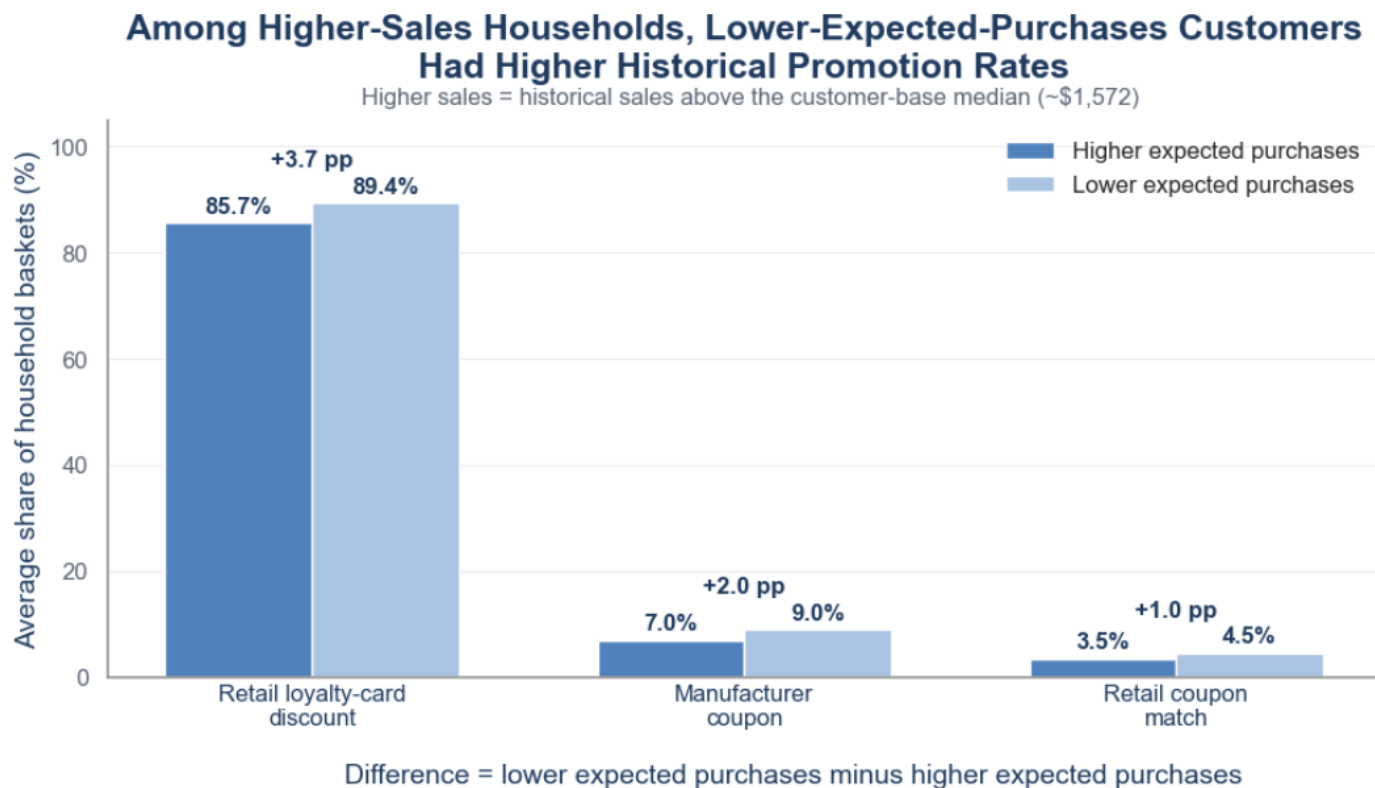
KEY TAKEAWAY

Because retailer discounts are widespread, **received vs. not received** does not separate customers well. More useful comparisons are **discount frequency, discount size relative to purchase value, and coupon activity across profiles**.

Historical sales and expected future activity define the customer profiles. Promotion patterns then help show how those profiles differ in their past response to or receipt of discounts and coupons.

Promotion Differences Are Most Consistent Among Higher-Sales Households

Higher-sales households are those with historical sales above the customer-base median (~\$1,572). Within this group, households with lower expected future activity historically received or redeemed promotions somewhat more often.



Higher-sales households (Above-median historical sales ~\$1,572) (shown in the figure)

- Lower-activity households received retail discounts more often: 89.4% vs. 85.7% of baskets (+3.7 percentage point).
- The same pattern appears for manufacturer coupons (9.0% vs. 7.0%) and coupon matches (4.5% vs. 3.5%).
- Manufacturer coupon value was also slightly higher: \$0.26 vs. \$0.18 per basket.

Lower-sales households (Below-median historical sales) (not shown in the figure)

- The main difference is retail-discount frequency: 82.9% vs. 78.4% (+4.5 percentage point).
- Coupon differences are small.

KEY TAKEAWAY:

- Higher future purchase activity was not accompanied by heavier historical promotion use. Among higher-sales households, the lower-expected-purchase group actually received or redeemed promotions somewhat more often.
- More promotion use does not necessarily signal stronger future purchases (**Descriptive pattern only, not causal**).

Future step: Check whether the pattern still holds after accounting for other customer differences/ factors.

Business implications: use the profiles to inform promotion and engagement next steps

The profiles help identify where different promotion and engagement approaches may be worth testing. **Note:** Promotion history can inform these next steps, but the analysis is descriptive. Other customer differences or factors may explain the patterns, so further modelling and controlled testing are needed to determine what actually works.

Profile	What the profile suggests	Possible next steps
Higher past sales / Higher expected purchases	Strong past contribution and higher expected future purchases. This is an important group to maintain, while avoiding unnecessary promotion cost.	This group had slightly lower historical promotion use than the higher-sales/lower-activity group , despite stronger expected purchase activity. Test whether more selective or personalized promotions can maintain purchasing without increasing promotion intensity.
Higher past sales / Lower expected purchases	Strong past contribution, but weaker expected future purchase activity. This may be a useful group to investigate for re-engagement.	This group already showed somewhat heavier historical discount and coupon use , so simply giving more promotions may not be the obvious answer. Test targeted re-engagement offers or communications against the current approach.
Lower past sales / Higher expected purchases	Lower past contribution, but higher expected future purchase activity. If that activity materializes, additional shopping occasions may create opportunities to grow sales contribution.	Promotion differences within the lower-sales group were small and mixed , so there is little evidence for a specific promotion strategy. Test basket-building, category expansion, cross-sell, or threshold offers and measure whether they increase sales per customer.
Lower past sales / Lower expected purchases	Lower contribution on both dimensions, although this is a large customer group.	Because promotion patterns do not clearly identify a stronger opportunity here, start with scalable, lower-cost engagement and test for responsive subgroups before using more costly incentives.

Key **limitations** of the analysis

- **Relative profiles:** “Higher” and “lower” are defined by customer-base medians, so the groups are practical comparison segments rather than natural customer clusters.
- **Predicted activity is uncertain:** Expected future purchases come from the BG/NBD model and should be interpreted as relative expected activity, not guaranteed future behavior.
- **Promotion findings are descriptive:** Other differences in sales, shopping frequency, category mix, or customer behavior may explain the observed promotion patterns.
- **No causal conclusion:** The analysis does not show that receiving or redeeming more promotions causes future activity to increase or decrease.
- **Purchases/ purchase activity is not customer value:** Historical sales measure past revenue and expected purchases measure future activity; neither directly measures profit or CLV.



References

Fader, P. S., Hardie, B. G., & Lee, K. L. (2005a). “Counting your customers” the easy way: An alternative to the Pareto/NBD model. *Marketing science*, 24(2), 275-284.

Fader, P. S., Hardie, B. G., & Lee, K. L. (2005b). RFM and CLV: Using iso-value curves for customer base analysis. *Journal of marketing research*, 42(4), 415-430.

Glossary / dictionary

Category	Term	Definition / Explanation
Data & business concepts	Household / Customer	One household in the dataset, treated as the customer unit throughout the analysis.
	Basket	One shopping trip or purchase occasion. Product-level transaction rows are combined into baskets.
	Historical period	The first 544 days of the data, used to calculate historical customer measures and fit the BG/NBD model. Purchases in the later holdout period are not used to fit the model or define the customer profiles.
	Historical sales	Total recorded sales value (\$) from a household’s purchases during the historical period. It reflects the retailer’s recorded sales after retailer-funded discounts. It is not the retailer’s product cost or profit, and it may differ from what the customer actually paid because manufacturer coupons are recorded separately.
	Historical contribution	A business term for how much recorded sales a household generated during the historical period. It does not represent profit, margin, or customer lifetime value.
	Actual future purchases	The number of baskets a household actually made during the 167-day holdout period . This is observed behaviour, unlike expected future purchases, which are model estimates.
	Expected future purchases	Model-estimated number of shopping baskets (purchase occasions) a household is expected to make over the next 167 days . Also referred to as expected future purchase activity or predicted future baskets . It does not represent expected future sales, profit, or CLV.
Customer profiling	Customer profile	A group defined by combining a household’s historical sales level with its expected future purchases.

	Higher / Lower	Relative to the customer-base median for that measure. These are comparison labels, not absolute judgments of customer quality or value.
	Aligned profile	A profile where historical sales and expected future purchases point in the same direction: higher/higher or lower/lower .
	Crossover profile	A profile where historical sales and expected future purchases point in different directions: higher sales/lower expected purchases or lower sales/higher expected purchases .
	Historical-sales quintile	Households divided into five approximately equal-sized groups based on historical sales, from Q1 = lowest historical sales to Q5 = highest historical sales . Used to compare future purchasing across different levels of past sales.
BG/NBD model concepts	BG/NBD	A repeat-purchase model for non-contractual settings. It uses purchase frequency, purchase timing, and observation time to estimate future purchase activity.
	Non-contractual purchasing environment	A setting where customers do not formally subscribe or cancel. Continued activity or dropout must therefore be inferred from observed purchasing behaviour.
	Purchase frequency (BG/NBD)	Number of repeat purchase occasions after the household's first purchase during the model-fitting period. In this project, it equals the number of historical baskets minus one.
	Recency (BG/NBD)	Number of days between a household's first and most recent purchase during its observed historical period. A household with only one purchase has a recency of 0.
	Observation time (BG/NBD)	Number of days from a household's first observed purchase to the end of the historical/model-fitting period . This can differ across households because they may make their first observed purchase at different times.
	Holdout period	A later 167-day period kept out of model fitting and used to compare predicted future purchases with actual future purchases.
	Historical-rate benchmark	A simple benchmark that projects a household's historical repeat-purchase rate forward into the holdout period. It provides a baseline against which BG/NBD can be evaluated.

Promotion concepts	Retail loyalty-card discount	A retailer-funded discount received on a basket.
	Manufacturer coupon	A manufacturer-funded coupon redeemed by the customer. It reduces what the customer pays, while the retailer is reimbursed by the manufacturer.
	Retail coupon match	An additional retailer-funded discount applied when a manufacturer coupon is matched.
	Promotion basket rate	The percentage of a household's baskets on which a specific promotion was received or redeemed.
	Average household promotion rate	The average of each household's individual promotion basket rate. Each household is weighted equally , regardless of how many baskets it made. For example, household rates of 90% and 70% produce an average household rate of 80%.
	Manufacturer coupon value per basket	Total manufacturer-coupon value redeemed by a household divided by all of its baskets , including baskets with no manufacturer coupon. It shows the average coupon value associated with each shopping occasion.
	Retail discount share of pre-discount value	Retail loyalty-card discount amount divided by the household's estimated purchase value before retailer-funded discounts . It shows how large the retailer discount is relative to purchase value. The denominator is an estimate and should not be interpreted as the exact shelf/list price.
	Percentage point (pp)	The absolute difference between two percentages. For example, 89% – 85% = 4 percentage points . This is different from a 4% relative change.
Model evaluation	MAE — Mean Absolute Error	Average absolute difference between predicted and actual basket counts across households. For example, an MAE of 11 means predictions are off by about 11 baskets per household on average , regardless of whether they are too high or too low. Lower is better.
	RMSE — Root Mean Squared Error	Measures prediction error like MAE, but gives greater weight to households with particularly large prediction errors. Lower is better.
	WAPE — Weighted Absolute Percentage Error	Total absolute prediction error divided by total actual baskets. It shows the overall magnitude of the errors relative to observed purchase volume. Lower is better.

	Aggregate % error	Compares total predicted baskets with total actual baskets across all households. A positive value means total purchases were overpredicted; a negative value means they were underpredicted; 0% means the predicted total matches the actual total.
	Mean bias	Average of predicted baskets – actual baskets across households. A positive value indicates average overprediction; a negative value indicates average underprediction. Values closer to 0 indicate less systematic over- or underprediction.
	Spearman correlation	Measures how similarly two variables rank households. Values closer to 1 indicate a stronger positive ranking relationship; values near 0 indicate little ranking relationship; values closer to -1 indicate opposite rankings. When comparing predicted with actual future purchases, a higher value means better ranking of households by future activity.